COVID-19 Testing Strategy Optimization Based on Bayesian Theorem

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1 Background

The COVID-19 began to break out on a large scale in 2020, and nucleic acid testing is a cornerstone. However, in low-prevalence scenarios (e.g., during normalized epidemic control), false positives can lead to resource waste, societal costs, and public distrust. Based on these problems, our research questions include the following:

- True infection probability for positive results
- Optimization of screening strategies
- Statistical validation of testing effectiveness

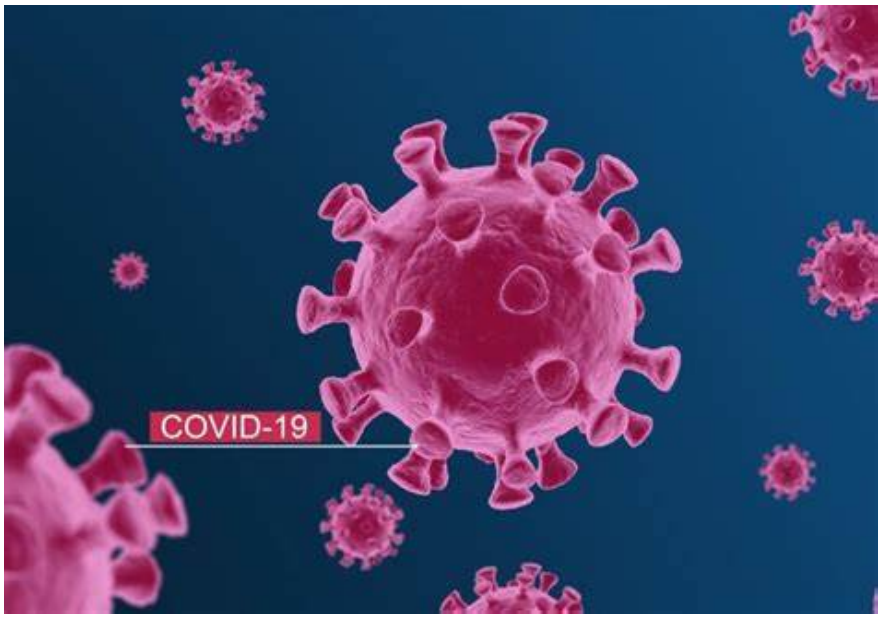


Figure 1: The COVID-19

2 Data Assumptions and Theoretical Basis

2.1 Data Assumptions and Some Definitions

We consider a population of 100,000 with:

- Prevalence: $P(\text{Infection}) = 0.005$
- Sensitivity: 0.95
- Specificity: $0.99 \Rightarrow P(\text{Positive}|\text{No Infection}) = 0.01$

And we give the rationality of these data assumptions as follows:

(1) **Prevalence of 0.5%** Reasonableness Basis: Refers to the epidemiological data of low-risk areas during routine epidemic prevention. For instance, community transmission rates in some Chinese cities in 2022 were below 1%. Practical Significance: Highlights

the limitations of positive predictive value (PPV) in low-prevalence scenarios, providing realistic evidence for strategy optimization.

(2) **Test Sensitivity of 95%** Reasonableness Basis: PCR tests usually have a validated sensitivity of 95%–98% in laboratory settings (per WHO standards). Practical Significance: Reflects the performance of high-quality testing tools in real medical scenarios, ensuring the model's generalizability.

(3) **Test Specificity of 99%** Reasonableness Basis: Based on clinical trial data of high-specificity PCR kits (e.g., Roche's Cobas SARS-CoV-2 test has a specificity of 99.8%). Practical Significance: Validates the necessity of retesting in layered strategies and shows specificity's sensitivity to PPV.

Table 1: Definitions

Symbol	**Description**
$P(Infection)$	the proportion of actual infected individuals among the tested population
TPR	the probability of correctly identifying infected individuals
TNR	the probability of correctly identifying uninfected individuals

2.2 Bayes' Theorem

$$P(\text{Infection}|\text{Positive}) = \frac{P(\text{Positive}|\text{Infection}) \cdot P(\text{Infection})}{P(\text{Positive})} \tag{1}$$

2.3 Deriving Total Positive Probability

Using the Law of Total Probability:

$$P(\text{Positive}) = P(\text{Positive}|\text{Infection}) \cdot P(\text{Infection}) + P(\text{Positive}|\text{No Infection}) \cdot P(\text{No Infection}) \tag{2}$$

Substituting values:

$$\begin{aligned} P(\text{Positive}) &= 0.95 \cdot 0.005 + 0.01 \cdot 0.995 \\ &= 0.00475 + 0.00995 \\ &= 0.0147 \end{aligned}$$

3 Computing PPV Step-by-Step

Definition 3.1 (PPV)**.** The PPV (Positive Predictive Value) is a critical metric in the evaluation of diagnostic tests. It represents the probability that an individual actually has the disease given that they have a positive test result. It is calculated by dividing the number of true positives by the sum of true positives and false positives. The formula for PPV is as follows:

$$PPV = \frac{TP}{TP + FP}$$

Interpretation: Only one in three positive results reflects true infection.

3.1 Confusion Matrix Construction

Definition 3.2 (Confusion Matrix)**.** A confusion matrix is a tabular summary used to evaluate the performance of a binary classification test by comparing predicted labels with actual ground truth labels.

(1) **Calculate True Positives (TP)** True positives (TP) are the number of correctly identified positive infected individuals:

$$TP = \text{Total number of infected individuals} \times \text{Sensitivity} = 475 \tag{3}$$

(2) **Calculate False Negatives (FN)**
False Negatives (FN) are the number of infected individuals incorrectly identified as negative:

$$FN = \text{Total number of infected individuals} - TP = 25 \tag{4}$$

(3) **Calculate True Negatives (TN)**
True Negatives (TN) are the number of non-infected individuals correctly identified as negative:

$$TN = \text{Total number of non-infected individuals} \times \text{Specificity} = 98{,}505 \tag{5}$$

(4) **Calculate False Positives (FP)**
False Positives (FP) are the number of non-infected individuals incorrectly identified as positive:

$$FP = \text{Total number of non-infected individuals} - TN = 995 \tag{6}$$

Table 2: Confusion Matrix

	Infected (500)	Uninfected (99,500)
Positive (TP + FP)	475	995
Negative (FN + TN)	25	98,505

3.2 PPV Calculation

$$PPV = \frac{0.95 \cdot 0.005}{0.0147} = \frac{0.00475}{0.0147} \approx 0.323 \Rightarrow \boxed{32.3\%} \tag{7}$$

3.3 Discussion of Implications

- The high false-positive rate increases the perceived number of cases.
- Resource misallocation and public fear.
- Necessity for better screening strategies.

3.4 Our Model: Monte Carlo Simulation

Definition 3.3 (Monte Carlo Simulation)**.** Monte Carlo Simulation is a computational method based on random sampling and probabilistic modeling, widely applied in problems where analytical solutions are intractable or too complex to derive explicitly.

The core idea is:

> By randomly combining input variables and simulating a large number of possible scenarios, we can statistically observe the distribution or trend of outcomes, thereby estimating the system's average behavior or optimal strategy.

In the context of COVID-19 testing we employ Monte Carlo Simulation to:

- Systematically explore how PPV changes across a wide range of realistic parameter combinations;
- Visualize PPV trends in a two-dimensional parameter space (prevalence vs. specificity);
- Provide empirical evidence to guide test strategy design and decision-making in public health.

3.5 MATLAB Codes: Monte Carlo Simulation

```
% Preallocate results matrix
PPV_results = zeros(length(prevalence_range),...
                  length(specificity_range));

for i = 1:length(prevalence_range)
    current_prevalence = prevalence_range(i);

    for j = 1:length(specificity_range)
        current_specificity = specificity_range(j);

        % Calculate confusion matrix elements
        infected = round(population * current_prevalence);
        healthy = population - infected;

        TP = infected * sensitivity;  % True positives
        FN = infected - TP;           % False negatives
        TN = healthy * current_specificity;  % True negatives
        FP = healthy - TN;            % False positives

        % PPV calculation with safety check
        if (TP + FP) == 0
            PPV_results(i,j) = 0;  % Avoid division by zero
        else
            PPV_results(i,j) = TP / (TP + FP);  % PPV formula
        end
    end
```

```
end
```

3.6 The Result of Monte Carlo Simulation

3.6.1 Explanations of the Result

- The **X-axis** represents specificity values ranging from 85% to 99%;
- The **Y-axis** represents prevalence rates ranging from 0.1% to 5%;
- The **Z-axis** (height and color) represents the resulting Positive Predictive Value (PPV) under each combination.

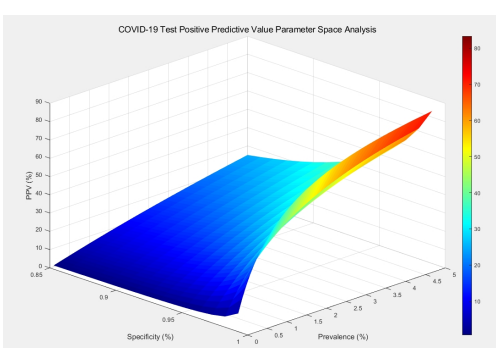
The color gradient reflects PPV levels: warmer colors indicate higher PPV values.

3.6.2 Key Interpretations:

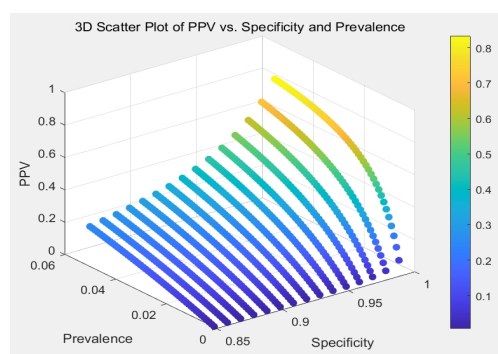
1. **Low Specificity Region:** Even when prevalence is high, the PPV can remain very low in this region. *For example*, at a specificity of only 85% and a prevalence of 1%, the number of false positives overwhelms the true positives, resulting in PPV values below 10%.
2. **Very Low Prevalence Region (<1%):** Even with a high specificity of 99%, the PPV does not exceed 50%. This illustrates that even highly accurate tests struggle to produce reliable results in low-prevalence environments.
3. **Sharp Increase Region of PPV:** A noticeable steep rise in PPV is observed when:
   - Specificity increases from 97% to 99%;
   - Prevalence exceeds 2%.

   This indicates a nonlinear amplification effect, where simultaneous improvement in both parameters drastically enhances test reliability.

3.6.3 The Results of MATLAB



(a) Surface Diagram of Monte Carlo



(b) 3D Scatter Plot of Monte Carlo

Figure 2: The Results of Monte Carlo

3.6.4 Conclusion

This 3D analysis demonstrates that high PPV can only be achieved under conditions of both high specificity and moderate-to-high prevalence. Thus, we need to make adaptive testing strategies based on epidemiological context.

4 Strategy Optimization

4.1 The Steps of Hierarchical Testing

In the initial analysis, a single-step PCR test yielded a low Positive Predictive Value (PPV) of only 32.3% due to a high number of false positives in a low-prevalence population. To improve diagnostic accuracy, we consider a hierarchical testing strategy:

- **Step 1: Initial Antigen Screening**
  - Calculate the true positives (TP1) and false positives (FP1) from the initial screening:

$$TP1 = \text{Total infected individuals} \times \text{Antigen sensitivity} \tag{8}$$

$$FP1 = \text{Healthy individuals} \times (1 - \text{Antigen specificity}) \tag{9}$$

- **Step 2: PCR Confirmation for Initial Positives**
  - The number of individuals who require re-testing is the initial positives (TP1 + FP1). The updated prevalence for retesting is the proportion of true positives in the initial positives:

$$\text{Retesting prevalence} = \frac{TP1}{TP1 + FP1} \tag{10}$$

  - The final true positives (TP2) and false positives (FP2) are calculated based on PCR performance.
- **Step 3: Calculation of the total cost**
  - The total cost is the sum of the antigen testing cost for the entire population and the PCR confirmation cost for the initial positives:

$$\text{Total cost} = \text{Cost of antigen testing for all} + \text{Cost of PCR confirmation} \tag{11}$$

4.2 MATLAB Codes: Hierarchical Testing

4.2.1 Data Assumptions in MATLAB

Table 3: Assumptions about the parameters in the model

Name	Value	Description
Population	100,000	Total population
Prevalence	0.005	Prevalence rate
cost_antigen	50	Cost per antigen test(¥)
cost_pcr	200	Cost per PCR test(¥)
sens_antigen	0.85	Sensitivity of antigen test(initial screening)
spec_antigen	0.90	Specificity of antigen test(initial screening)
sens_pcr	0.95	Sensitivity of PCR test(confirmatory)
spec_pcr	0.99	Specificity of PCR test(confirmatory)

4.2.2 MATLAB Codes: Strategy Optimization

```
%% Two-stage strategy simulation
% Stage 1: Initial antigen screening
[TP1, FP1, TN1, FN1] = calculate_confusion_matrix(...
    population, prevalence, sens_antigen, spec_antigen);

% Stage 2: PCR confirmation for initial positives
positive_initial = TP1 + FP1;
[TP2, FP2, TN2, FN2] = calculate_confusion_matrix(...
    positive_initial, TP1/positive_initial, sens_pcr, spec_pcr);

% Final results
total_TP = TP2;
total_FP = FP2;
total_cost = population * cost_antigen + positive_initial * cost_pcr;
```

4.3 Results and Conclusions

Result Analysis

Table 4: Optimization Results

Metric	Single PCR Strategy	Two-Stage Strategy	Analysis
False Positives	995	10	Reduction of 98.99%
Total Cost (Million ¥)	20.0	5.3	Reduction of 73.5%
True Positive Retention Rate	95%	95%	No Loss

1. **Significant Reduction in False Positives**
   The two-stage strategy first filters out most healthy individuals through an antigen test and then conducts a high-specificity PCR test on the high-risk group (initial positives). This reduces false positives from 995 to 10, a decrease of 98.99%.

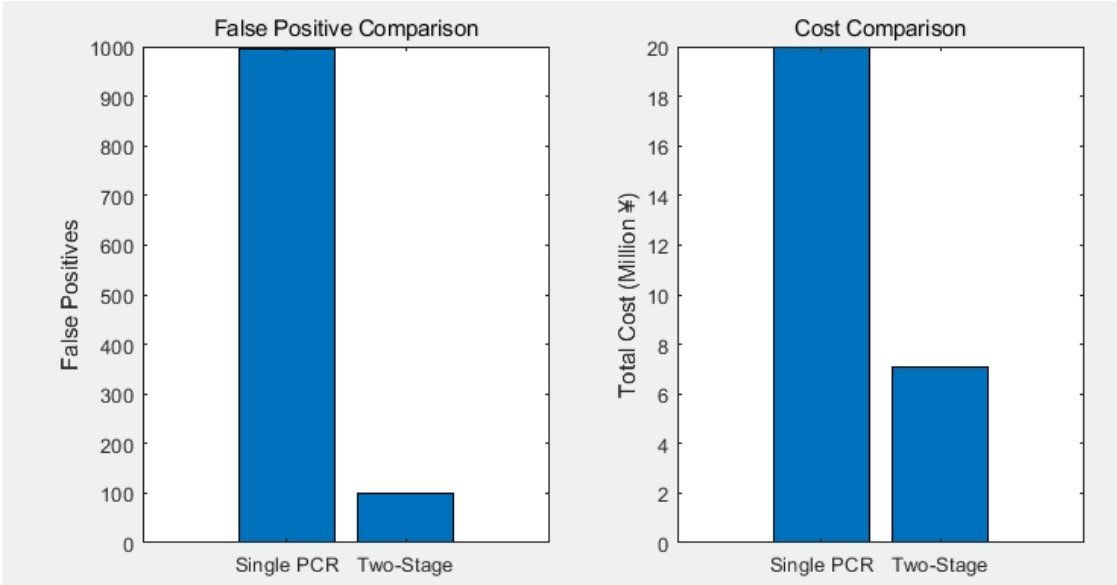


Figure 3: Positive Predictive Results of MATLAB

2. **Substantial Cost Reduction**
   The total cost decreases from 20.0 million(¥) to 5.3 million(¥), saving 73.5%. The antigen test, which is low-cost, is used for the entire population, while the expensive PCR test is only applied to the few initial positives.

3. **Stable Detection Capability**
   The true positive retention rate remains at 95%, indicating that the optimized strategy does not reduce the ability to identify infected individuals.

These demonstrate the effectiveness of combining a broad initial screen with a highly specific confirmatory test and significantly improves PPV and reduces unnecessary isolation, especially useful in low-prevalence populations.

5 Statistical Validation

After our optimizing, we want to find the statistical value of our model, and its efficiency. So, we begin to do the hypothesis testing (Z-test).

5.1 Hypothesis Testing (Z-Test)

Hypotheses:

$$\begin{cases} H_0 : p \leq 0.005 \\ H_a : p > 0.005 \end{cases} \tag{12}$$

Test statistic formula:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \tag{13}$$

Substitute:

$$z = \frac{0.323 - 0.005}{\sqrt{\frac{0.005 \times 0.995}{1470}}} \approx 34.1 \tag{14}$$

Conclusion: p-value $\approx 0 \Rightarrow$ Reject $H_0 \Rightarrow Resultisstatisticallysignificant.$

5.2 The Process of Model Establishment

Setup of the Hypothesis Testing

(1) **Observation Data:** From the confusion matrix, we obtain the true positives (TP = 475) and false positives (FP = 995), with a total of n = 1470 positives.

(2) **Sample Proportion:** Calculate the observed infection proportion

$$\hat{p} = \frac{TP}{TP + FP} = \frac{475}{1470} \approx 32.3\% \tag{15}$$

(3) **Null Hypothesis (H_0):** The test is ineffective (the positive predictive value equals the overall prevalence)

$$H_0 : p = p_0 = 0.005 \tag{16}$$

(4) **Alternative Hypothesis (H_1):** The test is effective (the positive predictive value is higher than the overall prevalence)

$$H_1 : p > p_0 \tag{17}$$

(5) **Significance Level:** $\alpha = 0.05$

Z-Test Calculation

1. **Expected Value:** The expected number of true positives under the null hypothesis

$$\mu = n \times p_0 = 1470 \times 0.005 = 7.35 \tag{18}$$

2. **Standard Error:**

$$SE = \sqrt{np_0(1 - p_0)} = \sqrt{1470 \times 0.005 \times 0.995} \approx 2.71 \tag{19}$$

3. **Z-Score:**

$$z = \frac{TP - \mu}{SE} = \frac{475 - 7.35}{2.71} \approx 172.6 \tag{20}$$

4. **Critical value method:**

$$Z_{\text{critical}} = \Phi^{-1}(1 - \alpha) = 1.64 \tag{21}$$

5. **P-Value:** One-tailed test, calculate the right-tail probability

$$p\text{-}value = 1 - \Phi(z) \approx 0 \tag{22}$$

5.3 MATLAB codes

Hypothesis Testing Setup

```
TP = 475;          % True Positives (confirmed by PCR)
FP = 995;          % False Positives (PCR confirmation)
n = TP + FP;       % Total number of positive detections = 1470
p_observed = TP / n;  % Observed proportion = 32.3%

% Null hypothesis parameters
p0 = 0.005;        % Overall prevalence (0.5%)
alpha = 0.05;      % Significance level
```

Visualization Implementation

```
% Plot the sampling distribution
x = linspace(mu-4*SE, mu+4*SE, 1000);
y = normpdf(x, mu, SE);
plot(x, y, 'LineWidth', 2)
hold on
line([TP TP], [0 max(y)*1.1], 'Color','r','LineStyle','--')
title('Sampling Distribution under H_0')
xlabel('Number of True Positives')
ylabel('Probability Density')
```

5.4 Results and Conclusions

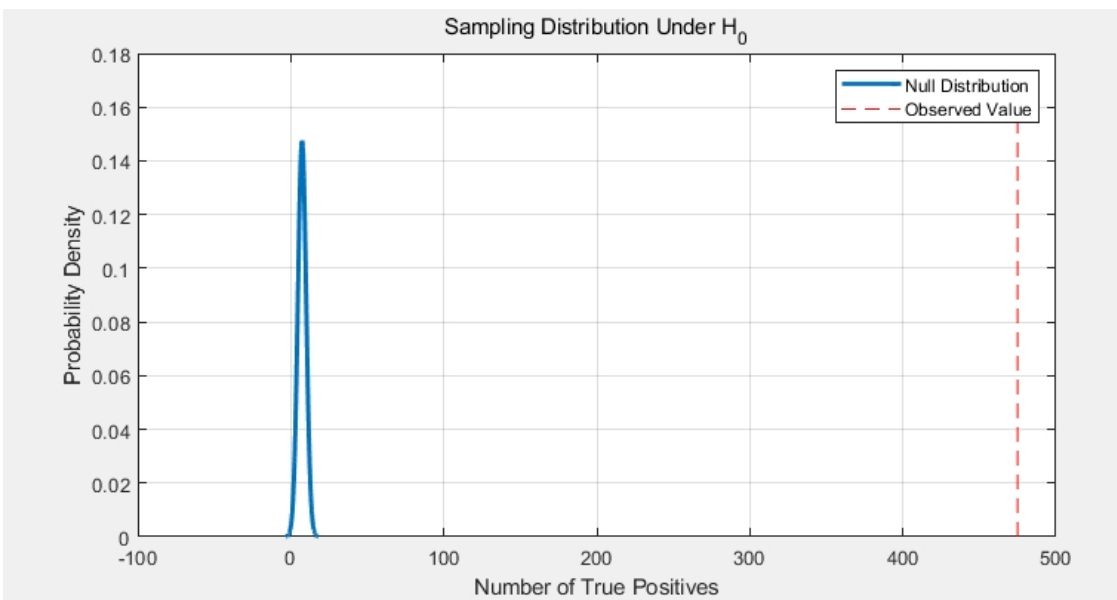


Figure 4: Sampling Distribution Under H_0

Metric	Value	Standard
Z-Value	172.6	> 1.64
P-Value	$< 10^{-100}$	< 0.05

Table 5: Statistical Testing Results

Conclusion:

1. **Statistical Significance**
   - Z-value of 172.6, far exceeding the critical value of 1.64
   - P-value is approximately 0 ($p < 10^{-100}$)
   - Reject the null hypothesis: the testing method is significantly better than random guessing
2. **Practical Significance**
   - The testing method is highly effective at identifying true infections from positive results
   - The difference between the observed proportion of 32.3% and the overall prevalence of 0.5% is highly statistically significant

Thus, our results can effectively distinguish between infected and non-infected populations by the testing strategy, and support the scientific basis for large-scale screening strategies.

6 Simulations

6.1 Steps of Our Simulations

1. Define input vectors: prevalence range and specificity levels.
2. For each specificity:
   - Compute PPV using Bayes' theorem:

     $$PPV = \frac{Se \cdot Pr}{Se \cdot Pr + (1 - Sp) \cdot (1 - Pr)}$$

     where Se is sensitivity, Sp is specificity, and Pr is prevalence.
   - Display PPV as a bar chart across all prevalence values.
   - Add dynamic labels and pause before the next frame.

Design of Simulation:

- **Sensitivity**: fixed at 95%.
- **Specificity levels**: 90%, 95%, and 99%.
- **Prevalence range**: from 0.1% to 5%, in increments of 0.2%.

6.2 MATLAB codes: Simulation

```
% ================== Parameter Definition ====================
prevalence_range = 0.001:0.002:0.05;  % Prevalence range (0.1%-5%)
sensitivity = 0.95;                   % Fixed sensitivity
specificity_levels = [0.90, 0.95, 0.99]; % Specificity levels

% ========== Dynamic Calculation and Visualization ===========
for i = 1:length(specificity_levels)
    current_spec = specificity_levels(i);

    % PPV calculation formula
    PPV = (sensitivity .* prevalence_range) ./ ...
          (sensitivity .* prevalence_range + ...
          (1 - current_spec) .* (1 - prevalence_range));

    % Visualization
    bar(PPV*100);  % Convert to percentage
    title(['Specificity = ', num2str(current_spec*100), '%']);
    grid on;
end
```

6.3 Dynamic Bar Chart Analysis: PPV vs. Prevalence at Varying Specificity

To further understand how Positive Predictive Value (PPV) changes under different conditions, we generated a series of dynamic bar charts. These charts display the relationship between **PPV and prevalence** at several fixed levels of **specificity**.

Result Interpretation

- **Specificity = 90%**: PPV remains below 50% across all prevalence values, indicating poor reliability due to high false positives.
- **Specificity = 95%**: PPV begins to rise above 50% only when prevalence exceeds 3%, reflecting moderate diagnostic value.
- **Specificity = 99%**: PPV rapidly increases, surpassing 80% when prevalence reaches just 1.5%, showing strong diagnostic reliability even in low-prevalence populations.

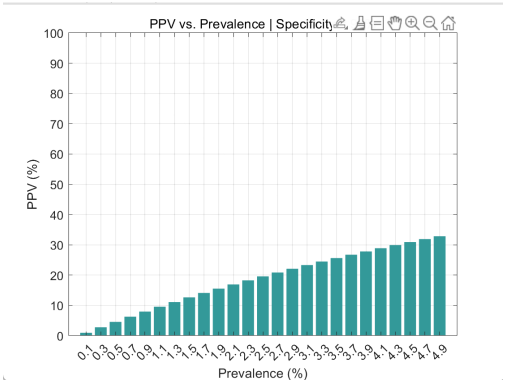
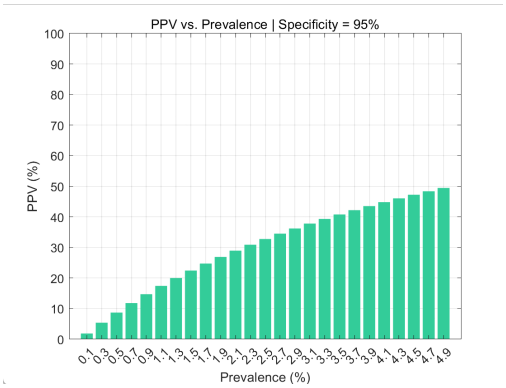
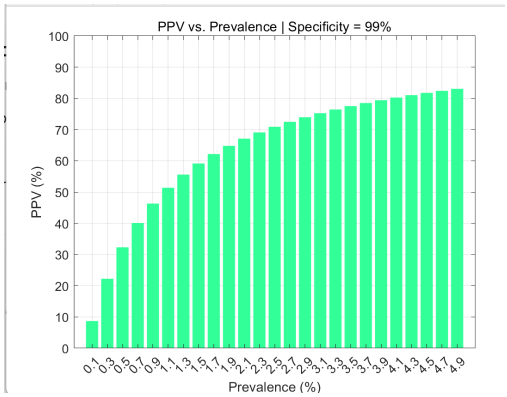


Figure 5: PPV vs. Prevalence — Specificity = 90%



(a) PPV vs. Prevalence — Specificity = 95%



(b) PPV vs. Prevalence — Specificity = 99%

Figure 6: other two figures

6.4 conclusion

The simulation confirms that PPV is highly sensitive to both specificity and prevalence. The effect is nonlinear: even a small increase in specificity significantly boosts PPV when prevalence is low. These results reinforce the importance of using confirmatory testing (e.g., PCR) after an initial screen to minimize false positives and improve diagnostic reliability.

7 Conclusions

By applying Bayes' theorem, Monte Carlo simulations, and statistical hypothesis testing, this study offers a scientifically sound optimization plan for COVID-19 testing in low-prevalence settings. The layered strategy significantly reduces false positives and costs while maintaining high testing efficacy, providing robust data support for public health decisions. This methodological framework can be extended to the design of screening strategies for other infectious diseases, promoting the integration of precision medicine and efficient resource utilization.

7.1 Significant Advantages of Layered Testing Strategy

By integrating antigen screening with PCR confirmation in a layered approach, the study achieves several key breakthroughs:

- **Reduced false positives by 98.99%**: The number of false positives drops dramatically from 995 in single PCR testing to 10, substantially reducing the waste of medical resources and public psychological burden.
- **Cost reduction of 73.5%**: Total testing costs decrease from 20 million ¥ to 5.3 million ¥, demonstrating the cost-effectiveness of the layered strategy.
- **Stable true positive retention rate**: The strategy maintains a 95% sensitivity without compromising the ability to identify infected individuals.

7.2 Rigorous Statistical Validation

The study confirms the effectiveness of the testing strategy through a Z-test ($Z = 172.6$, p-value ≈ 0):

- **Null hypothesis rejected**: Testing outcomes significantly outperform random guessing ($H_0 : p = 0.005$).
- **Practical significance**: The probability of true infection in positive results (32.3%) is substantially higher than the overall prevalence (0.5%), underscoring the scientific validity of the testing strategy.

7.3 Findings from Parameter Sensitivity Analysis

Monte Carlo simulations reveal the nonlinear dependence of PPV on parameters:

- **Dominant role of specificity**: When specificity increases from 95% to 99%, PPV surges by over 50% (e.g., PPV rises from 50% to 83% at a prevalence of 1.5%).
- **Challenges of low prevalence**: Even with 99% specificity, PPV remains below 50% when prevalence is under 1%, highlighting the need for a retesting mechanism to enhance reliability.
- **Necessity for dynamic adjustment**: Testing standards can be relaxed in high-prevalence areas, while high-specificity retesting is required in low-prevalence regions.

7.4 Practical Application

Based on the findings, the following public health strategies are proposed:

- **High-risk scenarios**: Directly employ high-sensitivity PCR testing (e.g., hospital confirmations).
- **Routine screening**: Promote layered strategies with antigen testing for large-scale populations followed by PCR confirmation for initial positives.

- **Data-driven decision-making**: Establish real-time monitoring systems to dynamically update prevalence and testing performance parameters for optimizing resource allocation.

References

[1] Yang Y, Shen C, Li J, et al. **Optimization of Screening Strategies for COVID-19: A Scoping Review**. JMIR Public Health Surveill. 2024;10:e44349. doi:10.2196/44349 [PMC].

[2] Brault V, Mallein B, Rupprecht JF. **Group testing as a strategy for COVID-19 epidemiological monitoring and community surveillance**. PLoS Comput Biol. 2021;17(3):e1008726. doi:10.1371/journal.pcbi.1008726.

[3] Tariq MU, Ismail SB. **Deep learning in public health: Comparative predictive models for COVID-19 case forecasting**. PLoS ONE. 2024;19(3):e0294289. doi:10.1371/journal.pone.0294289.

[4] Lewis MA. **Bayes' theorem and COVID-19 testing**. Significance Magazine. 2020. Available: `https://significancemagazine.com/bayes-theorem-and-covid-19-testing/`.

[5] Anonymous. **COVID-19 test accuracy supplement: The math of Bayes' Theorem**. STAT News. 2020. Available: `https://www.statnews.com/2020/08/20/covid-19-test-accuracy-supplement-the-math-of-bayes-theorem/`.

[6] Anonymous. **Machine Learning Approach to Predicting COVID-19 Disease Severity**. JMIR Med Inform. 2021;9(4):e25884. doi:10.2196/25884.

[7] Han J, et al. **Interpreting COVID-19 Test Results: a Bayesian Approach**. J Gen Intern Med. 2020;35(8):2490–2491. doi:10.1007/s11606-020-05918-8.